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Spatiotemporal Fourier neural operator-empowered super-resolution traffic flow field reconstruction from sparse observations☆

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ABSTRACT

An ideal traffic state reconstruction (TSR) framework should be able to generate traffic state values at any resolution level, thus achieving super-resolution. However, most existing studies are limited to fixed resolutions, which greatly limits the applicability of TSR methods in real-world scenarios. To overcome this bottleneck, this paper proposes a super-resolution TSR framework based on the spatiotemporal Fourier Neural Operator, named STFNO. The central concept of this framework is the development of mapping operators between functions within the frequency domain based on STFNO. This approach overcomes the constraints inherent in conventional pointwise function learning techniques, thereby facilitating the comprehension of fundamental universal principles governing traffic flow. The framework consists of two core tasks: (1) Resolution-fixed TSR: the model takes spatiotemporal coordinates as input and outputs traffic state values of the corresponding resolution; (2) Zero-shot super-resolution TSR: the model is trained with low-resolution traffic state data and then directly inferred based on high-resolution spatiotemporal coordinates to achieve zero-shot super-resolution reconstruction of traffic state. This paper conducts extensive experiments on multiple datasets at different spatiotemporal scales. In the resolution-fixed TSR task, STFNO effectively reconstructing traffic field evolution using sparse observations, with advantages in accuracy and a good balance between memory usage and computational efficiency. In the zero-shot super-resolution task, the powerful generalization capability of STFNO is fully demonstrated from three perspectives: temporal generalization, spatial generalization, and joint spatiotemporal generalization.

1. Introduction

Currently, artificial intelligence-based intelligent transportation system (ITS) is leading the new revolution in future mobility [1–3]. The advanced features of ITS, including real-time traffic control, dynamic guidance, congestion pricing, and advanced mobility services, rely heavily on comprehensive, accurate, and real-time traffic states data for decision-making [4,5]. Understanding the

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system states is crucial for the monitoring, analysis, and control of ITS systems [6]. The quality and coverage of traffic monitoring data directly determine the upper limit of the application efficiency of the ITS.

However, idealized full-area high-precision perception is almost impossible to achieve under current technological conditions. Limited by high construction costs, complex maintenance work, and physical environmental constraints, the deployment of existing traffic detection facilities is often sparse and unevenly distributed [7]. This results in the acquired traffic data being discrete in space and asynchronous in time, forming a large number of perception blind spots [8]. Therefore, there is an urgent need to accurately reconstruct the global high-resolution traffic state field from these limited, local observation points [9].

Scholars have conducted extensive research on traffic state reconstruction [10]. Traditional methods are primarily driven by mathematical and physical models, rely heavily on traffic flow theory [11]. Another alternative is early data-driven methods based on shallow feature extraction, including spatial sequences [12] or algebraic interpolation [13], and later the introduction of Bayesian statistics and other machine learning algorithms [14]. In recent years, end-to-end reconstruction methods have achieved impressive performance thanks to the powerful function approximation and feature extraction capabilities of deep learning [3,9,15,16]. These deep learning models possess powerful nonlinear mapping capabilities, enabling them to directly learn complex spatiotemporal dependencies from massive amounts of data [17,18]. Traffic flow is inherently a nonlinear dynamical system subject to strong spatiotemporal constraints. While its dynamic evolution is inherently uncertain, it adheres to fundamental physical laws [19]. End-to-end methods lack global regularization, necessitating the introduction of physical constraints across the entire spatiotemporal domain to improve the generalization and accuracy of reconstruction methods. The most representative approach is the physics-informed neural network (PINN) architecture [20].

However, despite the significant progress made in existing research, there is a common fundamental limitation: the vast majority of methods are constrained to fixed input and output resolutions, and they are usually trained and inferred on a predefined, uniform spatiotemporal grid. This means that once the model is trained, its output resolution is fixed. To obtain state fields of different resolutions, it is often necessary to tediously retrain the model or perform post-processing, which greatly limits the flexibility, adaptability and scalability of traffic state reconstruction in real-world applications. Macroscopic road network situation monitoring and microscopic local road section detailed analysis often require state fields of different resolutions as support. So, is there a technical framework that can fundamentally get rid of the dependence on fixed grids and truly achieve a ‘‘resolution-unlimited’’ traffic state reconstruction?

The Fourier neural operator (FNO) [21] has captured our attention precisely because it inherently decouples the model from the grid, offering a novel solution for overcoming resolution constraints. Unlike deep learning methods that learn mappings on a fixed, discrete spatiotemporal grid, FNO operates in the neural operator framework [22] and learns mappings between functions rather than between grid-specific tensors, which makes it particularly suitable for traffic state field reconstruction from sparse local observations and for inference across different spatiotemporal resolutions. Its core innovation lies in performing global convolution operations in the Fourier frequency domain, which enables it to efficiently capture complex spatiotemporal dependencies and global distribution patterns between functions with lower computational complexity. Its core concept is to learn the mapping from input functions to output functions. This means that once trained, the FNO model can directly accept spatiotemporal coordinates of arbitrary resolution as input and instantly infer the state values at the corresponding points, inherently endowed with superior resolution extrapolation and zero-shot super-resolution capabilities.

Therefore, to address the resolution constraints inherent in conventional traffic state field reconstruction methods, this study introduces a zero-shot super-resolution reconstruction framework grounded in the spatiotemporal Fourier Neural Operator, named STFNO. Within STFNO framework, we initially demonstrate the superior performance of STFNO in reconstructing traffic state fields when utilizing data at identical resolutions. Building on this validation, the model is subsequently trained exclusively on low-resolution traffic state data. Post-training, the model directly accepts high-resolution spatiotemporal information to generate high-precision, high-resolution traffic state fields, thereby enabling reconstruction that is unrestricted by resolution. The contributions of this paper are thus outlined as follows:

- We propose a traffic state field reconstruction method based on spatiotemporal Fourier neural operators, which establishes operator mappings between infinite-dimensional function spaces.
- We reformulate the learning of traffic flow patterns into a global modeling and learning task within the frequency domain through the application of the Fast Fourier Transform.
- We employ wave number truncation to preserve the principal frequency-domain components, while integrating low-rank decomposition to reduce model complexity.
- We conduct experiments on three real-world datasets to demonstrate that STFNO outperforms baseline models in both accuracy and generalization ability.

The remainder of this paper is organized as follows. Section 2 reviews related works. Section 3 describes the details of the model construction. In Section 4, we validate the proposed approach with three case study. Section 5 concludes our work and prospects.

2. Related literature review

This section reviews the related work on traffic state reconstruction and introduces the applications of neural operators in the traffic field.

2.1. Traffic state reconstruction

Related research mainly focuses on three types of methods: traffic flow model-driven, data-driven and data-model hybrid-driven. The core of the model-driven approaches is to use traffic flow models that describe traffic dynamics [23]. Such models are usually built based on empirical relationships. Among them, the LWR model is widely used due to its easy calibration and efficient solution [24]. For example, [25] applied the intrinsic property of the LWR model, the propagation speed of the shock wave, to reconstruct the traffic situation. In addition, high-order models are also used to estimate and reconstruct non-equilibrium traffic flow states [26]. Besides, early data-driven statistical and machine learning methods directly learned and extracted the shallow statistical characteristics and evolution laws of traffic flow from data [27,28]. For example, [14] proposed a distributed computing K-nearest neighbor section imputation method that maintains the geometric properties. [29] extended the Bayesian probability matrix decomposition model to high-order tensors for traffic data imputation. [30] developed a real-time freeway traffic state estimation method with online parameter updating for inhomogeneous traffic flow. [31] proposed a heterogeneity-guided tensor decomposition method to improve imputation performance under sparse and non-stationary data conditions.

However, the above methods have limitations in capturing the complex nonlinear spatiotemporal dependencies, which have prompted deep learning methods that can learn deep feature representations to become a new research focus [32]. For example, [33] integrated multiple processes such as identification, reconstruction, prediction and noise suppression in traffic density reconstruction into a unified framework. [34] proposed a pattern-adaptive generative adversarial network for TSR, showing that Bayesian inference enhances adaptability to varying traffic patterns. [35] proposed an integrated deep learning framework that combines structured deep network embedding with generative adversarial networks to reconstruct the traffic status. In order to reduce the reconstruction cost, PDEs have also been combined with deep learning models to improves accuracy and computational efficiency [36]. Given the inherent spatiotemporal characteristics of traffic flow fields, graph convolutional networks (GCNs) that can model spatial correlations in road networks and Transformers that capture temporal dependencies have become the current mainstream direction. [37] built a hierarchical graph convolutional architecture for macro and road layers, integrating multi-view spatiotemporal features to improve adaptability to different missing patterns.

Recently, research has shifted to data-model hybrid-driven approaches, among which PINN is particularly representative [38]. For example, [39] proposed an observer-informed deep learning model, which integrates a PDE-based observer with theoretical guarantees and a deep learning model to estimate traffic states from boundary data. [40] constructed a graph learning module and integrated it with a physics-informed module based on trapezoidal MFD to achieve traffic field interpolation and reconstruction.

However, despite these advancements, a common and critical challenge persists across most existing methods: their inherent dependency on a fixed, predefined spatiotemporal resolution. The vast majority of approaches are designed and trained for specific grid structures, limiting their flexibility and generalizability in real-world scenarios where multi-scale state reconstruction is often required. This resolution constraint not only affects practical applicability but also hinders the development of truly adaptive and scalable traffic monitoring systems.

2.2. Application of neural operators in traffic research

Neural operators are designed to learn mappings between infinite-dimensional function spaces [41]. A variety of neural operator architectures have been proposed, including deep operator networks (DeepONets), integral kernel neural operators, and Transformer-based neural operators [42]. The application of neural operators in transportation is still in its nascent stages, its potential for resolving fundamental challenges in spatiotemporal traffic modeling is significant. For example, [43] developed a physics-informed Fourier neural operator to learn solutions of macroscopic traffic flow models. Besides, our previous work adopted the physics-informed deep operator network (PI-DeepONet) to train a parameterized neural operator and integrate the physics of traffic flow into the operator learning process [44]. The powerful approximation ability of neural operators for partial differential equation systems also lays the foundation for their application in the more challenging field of traffic control. For example, [45] introduced neural operator to streamline the design of PDE boundary control for stabilizing stop-and-go traffic, achieving practical stability with improved computational efficiency. For ARZ models with uncertain parameters, a DeepONet-based adaptive boundary control is designed to learn the mapping from parameters to PDE backstepping kernels, improving stability and computational efficiency [46]. These explorations have initially demonstrated the significant potential of neural operators in traffic modeling and control, their application to the core problem of reconstructing spatiotemporal traffic state fields remains largely unexplored, leaving ample room for exploration. Importantly, existing traffic studies have predominantly adopted the DeepONet framework. From the perspective of spatiotemporal grid data processing, DeepONet and FNO differ fundamentally in both representation and computation. DeepONet usually encodes the input function through a branch network based on a finite set of sensor evaluations, while a trunk network is used to process the spatial–temporal query coordinates, thereby learning a function-to-point mapping. This design is flexible for irregularly sampled observations and pointwise prediction. However, when the task is to reconstruct a dense spatiotemporal field on regular grids, DeepONet does not explicitly exploit the structured tensor form of the grid and typically requires repeated coordinate queries for field-wide prediction. In contrast, FNO operates directly on grid-aligned field tensors and learns a function-to-function mapping over the entire spatiotemporal domain. By performing global spectral convolution in the Fourier domain, FNO can naturally capture long-range spatial–temporal dependencies and cross-scale patterns with high computational efficiency on regular grids. This property is highly compatible with the task of reconstructing continuous traffic state fields from sparse observations, particularly when the goal is to generate fields at arbitrary resolutions.

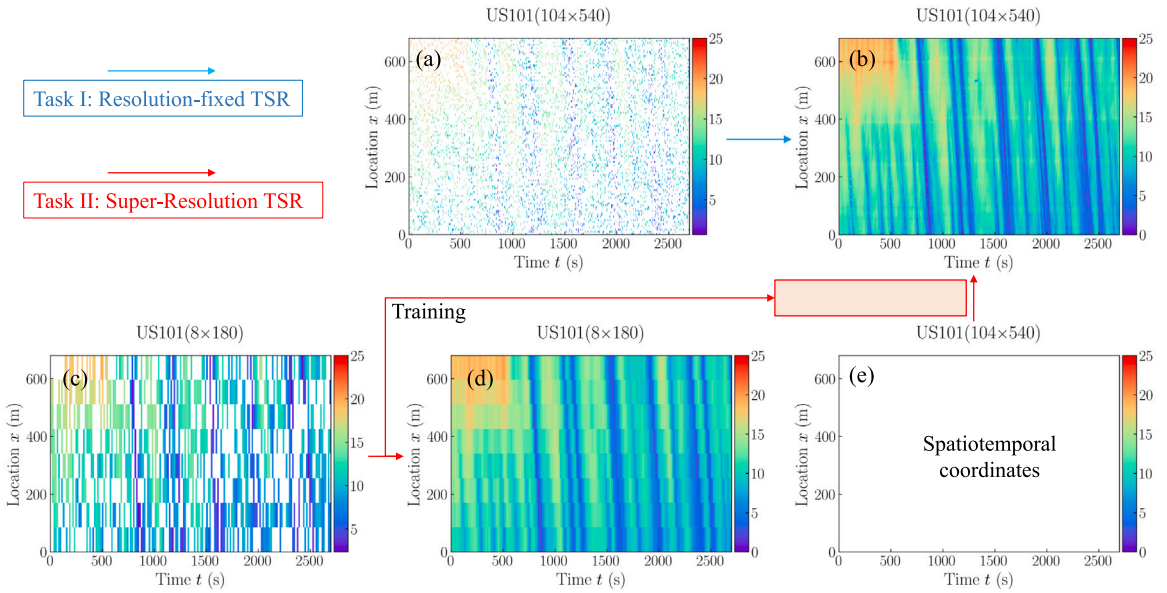


Fig. 1. Schematic diagram of the problem description of resolution-fixed TSR and zero-shot super-resolution TSR.

3. Methodology

3.1. Problem definition

This paper primarily solves two-stage tasks: traffic state reconstruction at the fixed resolution and zero-shot super-resolution reconstruction, as illustrated in Fig. 1 below. Traffic state reconstruction is defined as the process of restoring the traffic state data for the entire spatiotemporal field of a road section using partial observation data. Specifically, the study object is a traffic state variable, for a road segment of length L, where speed $v(x,t)$ is selected here. x and t represent the location and time of data collection, respectively. The complete spatiotemporal field is defined as Ω, the limited observation dataset is defined as subset $s_O \in s_\Omega$, and the mathematical description of the typical resolution-fixed TSR problem is defined as $s_O \to s_\Omega$, as shown in the process from Figs. 1(a) to 1(b) in Fig. 1 (blue arrow). On this basis, let S'_Ω be the high-resolution traffic status data in the entire space–time field, and $\mathcal{M}_O$ be the model trained based on low-resolution observation data. Then the zero-shot reconstruction problem can be defined as: $S'_\Omega = \mathcal{M}_O(\Omega_H)$, where Ω_H is the grid matrix corresponding to the high-resolution data, which only contains the space–time coordinates corresponding to the grid, that is, training is completed on low resolution data through the process from Figs. 1(c) to 1(d) in Fig. 1, and then combined with the refined spatiotemporal coordinates shown in Fig. 1(e) to reconstruct the high-resolution traffic state field shown in Fig. 1(b).

3.2. Overall architecture

Fig. 2 illustrates the proposed STFNO framework. To ensure close alignment with real-world scenarios, this paper assumes that the observation data is collected from sensors deployed at fixed locations at a specified frequency, representing a uniform grid. However, this data may contain random missing values due to sensor failures. Overall, the proposed framework consists of two steps: the classic resolution-fixed TSR step and the zero-shot super-resolution TSR step, which this paper focuses on. The former is actually a sub-step of the latter. In the resolution-fixed TSR step, we take a spatiotemporal grid as input. For each spatiotemporal grid, STFNO first uses a dimensionality promotion layer to upscale the spatiotemporal features to the corresponding dimension. This is then fed into the Fourier layer for processing. In each Fourier layer, the feature vector is processed using two channels, including a fast Fourier transform and a residual connection. The outputs are combined using a gated fusion mechanism to produce a feature vector, which is then passed to the next layer. After the Fourier layer, the feature vector is projected to the target dimension via a projection layer. Training supervision and model parameter updates are performed based on the observed data. After training, the model performs forward inference, producing restored global traffic state data. For the zero-shot TSR phase, this paper first completes the training and evaluation of the STFNO with low-resolution observation data. After obtaining the trained STFNO, high-resolution spatiotemporal grid features are directly input to obtain high-resolution traffic state data. Note that throughout the zero-shot reconstruction phase, high-resolution observation data is only used for subsequent performance evaluation. Therefore, the proposed framework does not require any high-resolution observation data in real-world applications. The following describes the various modules of the proposed framework in detail.

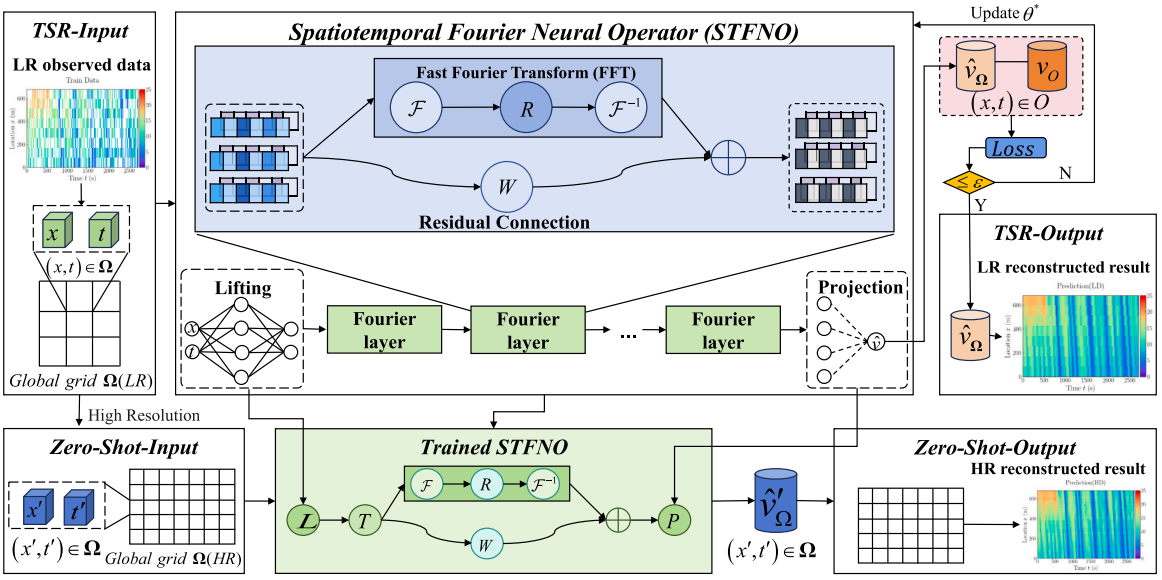


Fig. 2. Overall architecture of STFNO. The upper part is resolution-fixed TSR and the lower part is zero-shot super-resolution TSR.

3.3. Fourier neural operator for resolution-fixed TSR

This section mainly introduces the detailed process of using FNO to perform resolution-fixed TSR. Specifically, the introduced FNO is mainly used to learn the operator $\mathcal{G}((x,t),s(x,t))$ that maps the spatiotemporal coordinates to the traffic state. This model breaks through the limitations of traditional point-to-point mapping, enabling better generalization performance in unseen locations. Based on the data processing process, the model can be divided into three modules: the dimensionality enhancement layer, the Fourier layer, and the projection layer. These are described in detail below.

3.3.1. Dimension lifting layer

The role of the dimension lifting layer is to lift the input features to the dimension required by the Fourier layer. This goal is achieved through a fully connected layer. In this paper, the traffic state s is regarded as a function $s(x,t)$ related to time and space. Therefore, for any $s(x,t)$, the input features of FNO are the spatiotemporal coordinates (x,t), i.e., the number of feature channels $C_{in}=2$. Note that in FNO, the standard data format is a uniform grid, with each grid cell representing the features at the corresponding location. FNO utilizes tensor parallel processing. Specifically, for a grid of $n\times m$, the dimensionality boosting layer implements the following functions:

$$f_l : \mathbf{X} \to \mathbf{H} \tag{1}$$

where, $\mathbf{X}\in\mathbb{R}^{n\times m\times 2}$ is the initial feature tensor, $\mathbf{H}\in\mathbb{R}^{n\times m\times C}$ is the feature tensor after dimension enhancement, and C is the number of feature channels inside the Fourier layer.

3.3.2. Fourier layer

The Fourier layer is a core component of FNO. Its function is to map spatial data to the frequency domain through the Fourier transform, perform efficient linear transformations in the frequency domain, and then return it to the spatial domain through the inverse Fourier transform. Finally, it combines residual connections and activation functions to achieve nonlinear mapping. Because Fourier layers are often stacked in the same structure, they have the same input and output. For example, in the lth Fourier layer, after dimensionality enhancement, a multi-dimensional FFT is performed on the input $\mathbf{H}_l$ to achieve the spatial-to-frequency domain conversion:

$$\hat{\mathbf{H}}(k) = \mathcal{F}[\mathbf{H}(x,t)] \tag{2}$$

where $\hat{\mathbf{H}}\in\mathbb{R}^{n\times m\times C}$ is the transformed frequency domain tensor, $k=(k_1,\ldots,k_d)$ is the wavenumber vector, and k_i corresponds to the frequency component of the discrete Fourier transform in the ith dimension. It is worth noting that for any k_i, it must satisfy $k_i\in[-N_i/2, N_i/2-1]$ [47], and N_i is the number of discrete sampling points in the ith spatial dimension, corresponding to the resolution of the traffic data. Given that high-frequency components have little impact on the model results and may lead

to unnecessary waste of computational resources, FNO performs wavenumber truncation on the frequency domain tensor, retaining only K_i $(K_i \leq N_i)$ components:

$$\hat{\mathbf{H}}_l^{trunc}(k) = Trunc[\hat{\mathbf{H}}_l(k)] \tag{3}$$

where $Trunc(\cdot)$ represents the wavenumber truncation operator, $\hat{\mathbf{H}}_l^{trunc}(k) \in \mathbb{R}^{k_1 \times k_2 \times C}$ represents the frequency domain tensor after truncation, and k_1, k_2 are retained wavenumbers in the two dimensions. After wavenumber truncation, a complex linear transformation is performed on the characteristic channel of each wavenumber component to achieve parameter learning:

$$\hat{V}_l(k) = \hat{\mathbf{H}}_l^{trunc}(k) \cdot W_l^{(1)} \tag{4}$$

where $W_l^{(1)} \in \mathbb{R}^{C \times C}$ is the frequency domain weight matrix of the lth layer, and $\hat{V}_l(k) \in \mathbb{R}^{k_1 \times k_2 \times C}$ is the tensor after frequency domain transformation. After the frequency domain transformation, FNO provides an inverse fast Fourier transform to convert the frequency domain tensor into a spatial domain tensor:

$$V_l(x,t) = Re[\mathcal{F}^{-1}[\hat{V}_l(k)]] \tag{5}$$

where $\mathrm{Re}(\cdot)$ represents the real part, which preserves the positive frequency components in the frequency-domain tensor, and $V_l(x,t) \in \mathbb{R}^{n \times m \times C}$ is the spatial tensor after Fourier path mapping. Furthermore, due to wavenumber truncation in the previous step, only a portion of low-frequency wavenumbers is retained, causing the model to favor capturing global features while neglecting local ones. In real traffic systems, the evolution of traffic states is not purely local: queue propagation, congestion dissipation, and upstream–downstream interactions often produce long-range spatiotemporal dependencies across the monitored region. To address this, a spatial shortcut module is implemented in the Fourier layer, referencing the residual connection concept to incorporate local details. This design can also help alleviate the potential over-smoothing of sharp localized traffic transitions caused by wavenumber truncation, such as congestion boundaries and shockwave fronts. Specifically, this module implements the following functions:

$$S_l(x,t) = \mathbf{H}_l(x,t) \cdot W_l^{(2)} \tag{6}$$

where $S_l(x,t)$ is the spatial tensor after spatial shortcut mapping, and $W_l^{(2)} \in \mathbb{R}^{C \times C}$ is the spatial weight matrix of the lth layer. After obtaining the spatial tensors output by the two paths, the Fourier layer fuses the two through residual connections:

$$\mathbf{H}_{l+1}(x,t) = \sigma[V_l(x,t) + S_l(x,t)] \tag{7}$$

where $\sigma(\cdot)$ is the activation function used to implement nonlinear transformations. The above process describes the data processing of a single complete Fourier layer. Repeating it L times means stacking L Fourier layers.

3.3.3. Projection layer

The projection layer converts the stacked feature tensor into the target dimension. Specifically, after L Fourier layers, we can obtain the stacked feature tensor $\mathbf{H}_L(x,t)$. Since our output dimension is a single traffic state variable $s(x,t)$, the output dimension is $C_{out} = 1$, that is:

$$f_p : \mathbf{H}_L(x,t) \rightarrow \mathbf{S}(x,t) \tag{8}$$

where $\mathbf{S}(x,t) \in \mathbb{R}^{n \times m \times 1}$ is the target traffic state.

3.3.4. Loss function

It is important to emphasize that FNO uses complete uniform grids as samples during each training process, meaning that each training process processes a complete $n \times m$ grid. However, the fundamental reason for the success of the traffic state estimation task is the lack of observation data. Based on this concept, this paper uses a mask matrix to mark whether the corresponding grid has observation data:

$$\mathbf{mask} = \begin{cases} 1, & (x,t) \in O \\ 0, & else \end{cases} \tag{9}$$

On this basis, the loss function of FNO can be given as follows:

$$Loss = \frac{1}{N_\Omega} \sum_{i=1}^{N_\Omega} \left((\hat{s}(x,t) - s(x,t))^2 \times mask_i \right) \tag{10}$$

where N_Ω is the number of grid points, and $mask_i$ is the mask value of the ith point. Note that this process means that only observation points participate in loss calculation and reverse parameter updates throughout the training process, completely avoiding the risk of data leakage and ensuring the reliability of the proposed process in practical applications.

Based on the above content, the complete pseudocode for the resolution-fixed TSR can be provided as pseudocode 1:

Algorithm 1 STFNO-Based Resolution-Fixed Traffic State Reconstruction

Require: Dataset D (complete spatiotemporal state $s_{\mathcal{G}}$, observed subset $s_{\mathcal{O}}$), mask matrix $\mathcal{M}$ ($\mathcal{M}_i = 1$: observed grid; $\mathcal{M}_i = 0$: missing grid), FNO params θ_{FNO} (W_{up}: dim-elevation weight; $\{W^l_{\text{freq}}, W^l_{\text{space}}\}^L_{l=1}$: Fourier layer weights; W_{proj}: projection weight), Fourier config (L: stacked layers; k_1, k_2: wavenumber truncation), EPOCHS, learning rate η.

Output: Reconstructed spatiotemporal state $\hat{s}_{\mathcal{G}}$, converged θ^*_{FNO}

Initialize: θ_{FNO} (random initialization)
for epoch ∈ EPOCHS **do**
  Extract spatiotemporal coordinate tensor $X \in \mathbb{R}^{N_x \times N_y \times 2}$ from D (N_x: spatial grids; N_y: temporal grids);
  Dimension Elevation: $X_{\text{up}} = W_{\text{up}} \cdot X$ (fully-connected layer);
  for $l = 1$ to L **do**
    $X_{\text{freq}} = \text{FFT}(X_{\text{up}})$;
    $X_{\text{freq,cut}} = \text{Truncate}(X_{\text{freq}}, k_1, k_2)$;
    $X_{\text{freq,trans}} = W^l_{\text{freq}} \cdot X_{\text{freq,cut}}$;
    $X_{\text{space}} = \text{Re}(\text{IFT}(X_{\text{freq,trans}}))$;
    Spatial Shortcut: $X_{\text{shortcut}} = W^l_{\text{space}} \cdot X_{\text{up}}$;
    $X_{\text{fuse}} = \text{ReLU}(X_{\text{space}} + X_{\text{shortcut}})$;
    $X_{\text{up}} = X_{\text{fuse}}$;
  end for
  Projection: $\hat{s} = W_{\text{proj}} \cdot X_{\text{up}}$;
  Data Loss: $\mathcal{L}_{\text{data}} = \frac{1}{\sum_i \mathcal{M}_i} \sum_i \mathcal{M}_i \cdot (s_{\mathcal{G},i} - \hat{s}_i)^2$ (observed points only);
  Update θ_{FNO}: $\theta_{\text{FNO}} \leftarrow \theta_{\text{FNO}} - \eta \cdot \nabla_{\theta_{\text{FNO}}} \mathcal{L}_{\text{data}}$;
end for
Inference: Input X to θ^*_{FNO}, repeat Steps 3–9, output $\hat{s}_{\mathcal{G}}$;
return $\hat{s}_{\mathcal{G}}, \theta^*_{\text{FNO}}$

3.4. Zero-shot super-resolution TSR

As mentioned above, zero-shot is based on resolution-fixed TSR, but does not rely on any high-resolution observational data. This process can be divided into three sub-steps: low-resolution training, high-resolution spatiotemporal grid construction, and high-resolution result inference. These steps are described in detail below.

3.4.1. Low-resolution training

The purpose of this module is to provide a pre-trained model for zero-shot. Compared with Section 3.3, the core difference of this section is that during the training process, we use the feature tensor $\mathbf{X}_{LD} \in \mathbb{R}^{N_{LD} \times M_{LD} \times 2}$ corresponding to the low-resolution grid as input and supervise the model training with the low-resolution observation dataset $\mathcal{D}^{LD}_O$, that is:

$$\theta * = argmin_\theta \; L_{\mathcal{D}^{LD}_O}(\mathcal{M}_\theta(\cdot)) \tag{11}$$

where θ^* is the optimized model parameter, $L_{\mathcal{D}^{LD}_O}$ is the loss function based on low-resolution observation data $\mathcal{D}^{LD}_O$, and $\mathcal{M}_\theta$ is the parameterized model. After training is complete, the corresponding model parameters are saved for later use.

It should be noted that due to the low spatiotemporal resolution of low-resolution data, special consideration needs to be given to the retention range of frequency domain components. Specifically, for a certain dimension, assuming the number of sampling points in the input data is N, after converting to the frequency domain through FFT, N frequency domain modes will be obtained, with an index range of $[0, N-1]$. The corresponding physical meanings are as follows:

Index $k \in [0, N/2]$: corresponds to positive frequencies, with the frequency magnitude increasing as k increases, and the maximum positive frequency corresponding to index $k_{max} = N/2$;

Index $k \in [N/2+1, N-1]$: corresponds to negative frequencies, which are symmetric mirrors of positive frequencies, with no additional information.

According to the Nyquist theorem, the sampling frequency $f_s = N$ of the input data determines the highest effective frequency as:

$$f_e = N/2 \tag{12}$$

Please note that, according to f_e, the maximum effective frequency domain index k_{max} can be determined, that is

$$k_{max} = N/2 \tag{13}$$

In other words, modes exceeding this index are either negative frequencies or aliased high frequencies, which are considered invalid components. For STFNO, n_modes (n_x, n_t) represents the number of frequency domain modes retained in each dimension,

which corresponds to the maximum value of the frequency domain index mentioned earlier. Therefore, during the training process, both spatial and temporal dimensions must satisfy the following condition:

$$\begin{aligned} n_x &\le N_x/2 \\ n_t &\le N_t/2 \end{aligned} \tag{14}$$

In the formula, N_x and N_t represent the number of spatial and temporal sampling points, respectively, which signifies the resolution of the dataset in the spatial and temporal dimensions. It should be noted that during the low-resolution training process described here, due to significant differences in the spatial and temporal dimensions of various datasets, and given that this process is essentially a pre-training step, the variations in computational cost usually do not impact practical deployment applications. Based on this, in the pre-training process, in order to ensure the generalization performance of STFNO as much as possible, the number of modes in the spatial and temporal dimensions is determined by dividing the resolution by 2 and then rounding down, that is:

$$\begin{aligned} n_x &= [N_x/2] \\ n_t &= [N_t/2] \end{aligned} \tag{15}$$

In the formula, $[x]$ represents the largest integer not exceeding x.

3.4.2. High-resolution spatiotemporal grid construction

The purpose of this module is to construct a spatiotemporal grid with the same area and time range as the low-resolution grid but with higher spatial resolution, thus providing input for zero-shot. Specifically, in the zero-shot task referred to in this article, the spatiotemporal dimensions of the feature tensor $\mathbf{X}_{HD} \in \mathbb{R}^{N_{HD} \times M_{HD} \times 2}$ of the high-resolution grid satisfy:

$$\begin{aligned} N_{HD} &= N_{LD} \cdot R_x \\ M_{HD} &= M_{LD} \cdot R_t \end{aligned} \tag{16}$$

where N_{HD}, M_{HD} are the spatial and temporal resolutions corresponding to $\mathbf{X}_{HD}$, and R_x, R_t represents the corresponding magnification. Note that since the features used in this paper are two-dimensional coordinates (x, t) of space and time, high-precision feature tensors can be generated based on actual application requirements without providing additional data.

3.4.3. High-resolution result inference

After completing the above preparation process, input $\mathbf{X}_{HD}$ into the pre-trained model $\mathcal{M}_\theta$ for direct inference:

$$\mathbf{S}_{\Omega}^{HD}(x,t) = \mathcal{M}_\theta\left(\mathbf{X}_{HD}(x,t)\right) \tag{17}$$

where $\mathbf{S}_{\Omega}^{HD}(x,t) \in \mathbb{R}^{N_{HD} \times M_{HD} \times 1}$ is the high-precision traffic status data obtained by inference of the pre-training model.

Therefore, the inferred high-resolution traffic field is not obtained from additional high-resolution observations, but from the cross-resolution representation of the traffic state field learned by STFNO.

Based on the above content, the complete pseudocode for the zero-shot super-resolution TSR can be provided as pseudocode 2:

Algorithm 2 STFNO-Based super-resolution Traffic State Reconstruction

Require: Low-res dataset $\mathcal{D}_{\text{low}}$ ($s_{\text{low},\mathcal{G}}$, $s_{\text{low},\mathcal{O}}$, $\mathcal{M}_{\text{low}}$), initial $\theta_{\text{FNO,init}}$, Fourier config (L, k_1, k_2), $\text{EPOCHS}_{\text{low}}$, η_{low}, high-res magnifications (R_x: spatial; R_t: temporal), low-res grid dims $N_{x,\text{low}} \times N_{y,\text{low}}$.

Output: High-res spatiotemporal state $\hat{s}_{\text{high},\mathcal{G}}$

Step 1: Low-Resolution Pre-Training

Call Alg. 1 (FNO-RF-TSR) with $\mathcal{D}_{\text{low}}, \mathcal{M}_{\text{low}}, \theta_{\text{FNO,init}}, L, k_1, k_2, \text{EPOCHS}_{\text{low}}, \eta_{\text{low}}$;
Obtain converged θ^*_{FNO};

Step 2: High-Resolution Grid Construction

High-res dims: $N_{x,\text{high}} = R_x \cdot N_{x,\text{low}}$, $N_{y,\text{high}} = R_t \cdot N_{y,\text{low}}$;

Generate high-res coordinate tensor $X_{\text{high}} \in \mathbb{R}^{N_{x,\text{high}} \times N_{y,\text{high}} \times 2}$;

Step 3: High-Resolution Inference

$X_{\text{high,up}} = W^*_{\text{up}} \cdot X_{\text{high}}$;

for $l = 1$ to L **do**

  $X_{\text{high,freq}} = \text{FFT}(X_{\text{high,up}})$;
  $X_{\text{high,freq,cut}} = \text{Truncate}(X_{\text{high,freq}}, k_1, k_2)$;
  $X_{\text{high,freq,trans}} = W^{*l}_{\text{freq}} \cdot X_{\text{high,freq,cut}}$;
  $X_{\text{high,space}} = \text{Re}(\text{IFT}(X_{\text{high,freq,trans}}))$;
  $X_{\text{high,shortcut}} = W^{*l}_{\text{space}} \cdot X_{\text{high,up}}$;
  $X_{\text{high,fuse}} = \text{ReLU}(X_{\text{high,space}} + X_{\text{high,shortcut}})$;
  $X_{\text{high,up}} = X_{\text{high,fuse}}$;

end for

$\hat{s}_{\text{high},\mathcal{G}} = W^*_{\text{proj}} \cdot X_{\text{high,up}}$;

return $\hat{s}_{\text{high},\mathcal{G}}$

4. Experiment

The experiment is divided into two parts, **resolution-fixed TSR** and **zero-shot super-resolution TSR**. The performance verification experiments were conducted on three specific cases: (1) the NGSIM US 101 freeway dataset; (2) the NGSIM I80 freeway dataset; (3) the Tokyo expressway dataset. The purpose of selecting three data sets is to more fully test the generalization advantages of FNO in traffic field reconstruction. Three types of zero-shot experiments, namely temporal generalization, spatial generalization and spatiotemporal generalization, are carried out in turn.

4.1. Dataset description

NGSIM US 101 freeway dataset. This dataset records the position and movements of each vehicle using camera video within a monitoring area of approximately 680 m and 2770 s, as shown in Fig. 3(a). We use this dataset to generate simulated fixed detector data [38]. Specifically, a distance of 20 ft and a time interval of 5 s were selected, during which we averaged the instantaneous speeds of all vehicles passing through the detector to obtain speed. After processing, valid grids with a spatiotemporal dimension of 104X540 were finally obtained, as shown in Fig. 3(b).

NGSIM I80 freeway dataset. This dataset, along with the US 101 dataset, is part of the NGSIM dataset project. It records the position and movements of each vehicle using camera video within a monitoring area of approximately 500 m and 2700 s, as shown in Fig. 3(c). Similar to the US 101 dataset, the original I80 dataset was gridded and an 81X360 velocity field was obtained, as shown in Fig. 3(d).

Tokyo Metropolitan expressway dataset [48]. This dataset is sourced from the loop detector of the Tokyo Metropolitan expressway corridor, as shown in Fig. 3(e). The two-lane-long corridor is about 13.3 km, with 23 loop detectors spaced at intervals of 210 m to 1030 m. The loop detector data, collected from August 11 to August 17, 2018, includes flow and speed measurements every 5 min. The global state of speed are shown in Fig. 3(f), which describe congestion patterns (stop and go waves caused by sag sections).

4.2. Experiment setup

The mean absolute error (MAE) and root mean square error (RMSE) are selected to evaluate the performance of STFNO model:

$$\begin{aligned} MAE &= \frac{1}{M}\sum_{i=1}^{M}\left|\hat{s}_i - s_i\right| \\ RMSE &= \sqrt{\frac{1}{M}\sum_{i=1}^{M}\left(\hat{s}_i - s_i\right)^2} \end{aligned} \tag{18}$$

where $\hat{s}_i$ represents reconstructed speed, $\hat{s}_i$ represents the ground truth, M is the number of test samples.

The baselines we have selected include NN, LSTM, Transformer and TimeMixer [49]. Since TimeMixer needs to be modified into a point-to-point version, it will be referred to as TimeMixerPP from now on. All models are executed on an Intel(R) Core(TM) i9-14900K CPU@3.20 GHz and an NVIDIA RTX 4090 GPU, using the Pytorch framework. We set 10%, 20%, 30%, 40% and 50% as the proportion of observable data in sparse detection data scenarios.

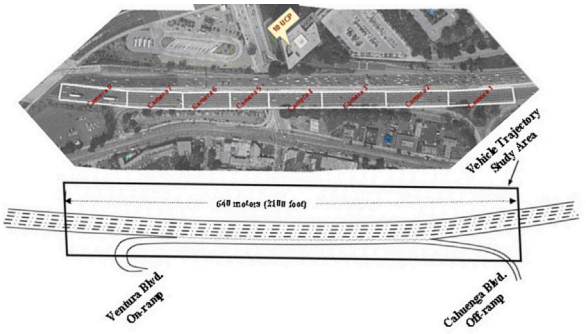
In STFNO, the number of Fourier layer is 4, the hidden layer dimension is 32, the activation function is Gaussian Error Linear Unit (GELU). A unique parameter in STFNO, the modes dimension, corresponds to the number of modes stored in different dimensions. Here, it corresponds to the spatial and temporal dimensions, set to 12 and 16, respectively. Nyquist Sampling Theorem [47] is employed here, and in subsequent zero-shot super-resolution experiments, the number of modes in the spatiotemporal dimensions is calculated by dividing the resolution by two and rounding down. Besides, dropout is set to 0.01 and the maximum number of epochs is 40000.

4.3. Results: Resolution-fixed TSR

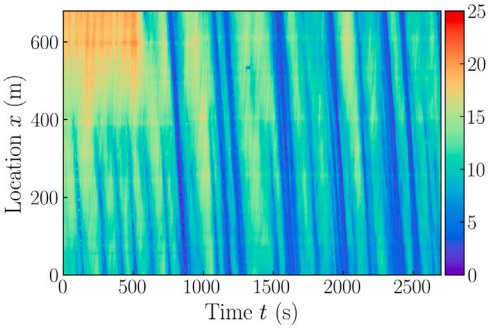
4.3.1. Accuracy of STFNO and baseline models

Table 1 presents the speed reconstruction results for the Tokyo case across sparsity levels from 10% to 50%, highlighting the strong performance of STFNO. At 10% sparsity, STFNO achieves a MAE of 3.92, establishing a solid foundation, with competitive RMSE. Moving to 20% sparsity, STFNO improves further, showing a clear downward trend in both MAE and RMSE compared to baselines. This trend continues at 30% sparsity, where STFNO achieves a MAE of 3.04 and an RMSE of 5.07. When the sparsity reaches 40% and 50%, the MAE of STFNO decreases to 2.86 and 2.82, respectively, and the RMSE is 4.69 and 4.59, respectively. Although STFNO does not achieve the lowest errors, its results remain close to the TimeMixerPP across all sparse scenarios, which verifies the effectiveness of STFNO.

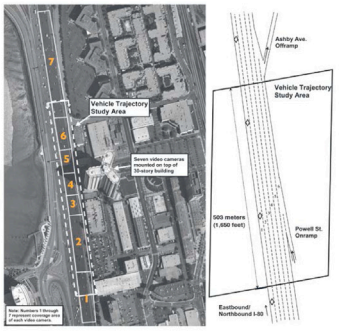
Table 2 shows the speed reconstruction results of STFNO and the baseline model on the NGSIM I80 case. Relying on 10% available observation data for model training and testing, STFNO achieved a MAE of 2.82 and an RMSE of 3.95, slightly lagging behind the Transformer and TimeMixerPP. As the available data ratio increases from 20% to 50%, STFNO shows a steady improvement trend. At 20% sparsity, STFNO achieves a MAE of 2.26 and an RMSE of 3.17, and this trend continues at 30% sparsity, where the MAE and RMSE further decrease to 2.01 and 2.81, respectively. At 40% and 50% sparsity, STFNO continues to deliver stable reconstruction results, with a MAE of 1.92 and an RMSE of 2.74 at 40% sparsity; at 50% sparsity, the MAE is 1.83 and the RMSE is 2.62. Overall, STFNO remains competitive on the NGSIM I80 dataset and demonstrates effective fixed-resolution speed field reconstruction performance across different sparse scenarios.



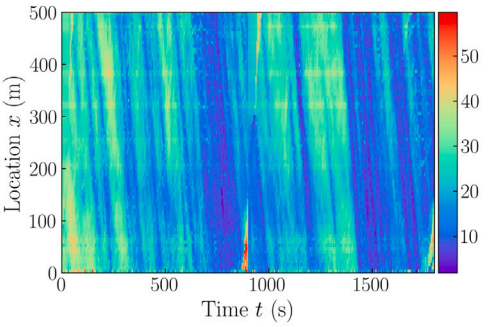
(a) Study area of US 101 dataset



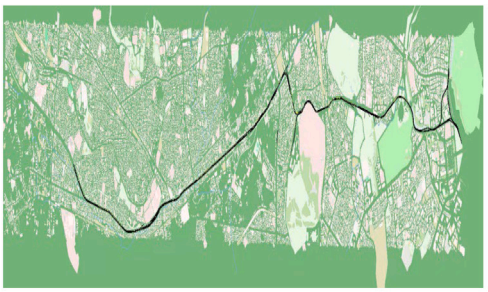
(b) Global speed field of the US 101 dataset



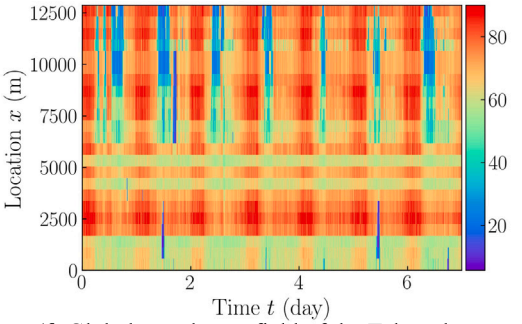
(c) Study area of I-80 dataset



(d) Global speed state field of the I-80 dataset



(e) Study corridor of Tokyo dataset



(f) Global speed state field of the Tokyo dataset

Fig. 3. Overview of experimental datasets. (a) and (b) are NGSIM US 101 freeway dataset, (c) and (d) are NGSIM I80 freeway dataset, (e) and (f) are Tokyo expressway dataset.

Table 1
Speed reconstruction comparison of proposed STFNO and baselines under various sparse scenarios in Tokyo case.

Model	10%		20%		30%		40%		50%	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
NN	6.11	9.69	5.72	9.27	4.18	6.71	4.16	6.64	4.03	6.26
LSTM	4.56	7.73	3.93	6.59	3.90	6.42	3.48	5.68	3.46	5.63
Transformer	3.96	7.06	3.80	6.60	3.54	6.22	3.20	5.40	3.03	5.18
TimeMixerPP	3.45	6.10	3.15	5.40	2.93	4.91	2.78	4.57	2.67	4.37
STFNO	3.92	7.28	3.26	5.57	3.04	5.07	2.86	4.69	2.82	4.59

Table 3 presents the speed reconstruction results for the NGSIM US101 case, shifting to a higher spatiotemporal resolution of 104×540. STFNO shows highly competitive performance across all sparse scenarios in this case. With 10% available detection data, STFNO achieves an MAE of 0.55 and RMSE of 0.75, which is comparable to the TimeMixerPP. This competitive performance persists as the proportion of available data increases. When the available detection data ratio reaches 50%, the MAE and RMSE of STFNO are 0.39 and 0.53 respectively, remaining close to the strongest TimeMixerPP. These results underscore the competitiveness of STFNO

Table 2
Speed reconstruction comparison of proposed STFNO and baselines under various sparse scenarios in NGSIM I80 case.

Model	10%		20%		30%		40%		50%	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
NN	2.79	3.70	2.60	3.44	2.53	3.41	2.30	3.10	2.15	2.87
LSTM	2.78	3.70	2.48	3.35	2.20	3.00	2.04	2.80	2.01	2.74
Transformer	2.70	3.64	2.38	3.25	2.16	3.02	2.13	3.00	2.03	2.86
TimeMixerPP	2.38	3.25	1.98	2.79	1.84	2.58	1.70	2.42	1.66	2.36
STFNO	2.82	3.95	2.26	3.17	2.01	2.81	1.92	2.74	1.83	2.62

Table 3
Speed reconstruction comparison of proposed STFNO and baselines under various sparse scenarios in NGSIM US101 case.

Model	10%		20%		30%		40%		50%	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
NN	1.25	1.68	0.73	0.97	0.66	0.87	0.65	0.84	0.64	0.84
LSTM	0.97	1.35	0.75	1.02	0.61	0.84	0.60	0.81	0.49	0.67
Transformer	0.73	1.03	0.58	0.80	0.52	0.71	0.51	0.70	0.45	0.62
TimeMixerPP	0.55	0.75	0.45	0.63	0.41	0.57	0.38	0.53	0.36	0.51
STFNO	0.55	0.75	0.44	0.60	0.41	0.57	0.40	0.54	0.39	0.53

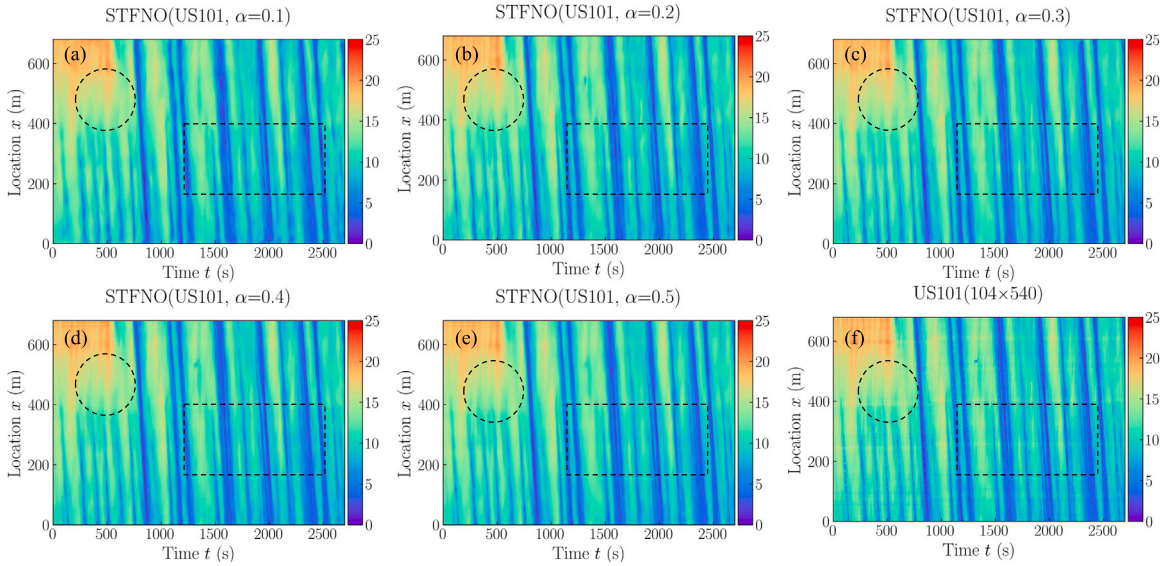


Fig. 4. Heatmap of speed field reconstruction by STFNO under different sparse scenarios. (a)–(e) correspond to available data rates of 10%, 20%, 30%, 40%, and 50%, respectively. (f) is the global observed speed field.

in handling complex, high-resolution traffic data under sparse conditions. This competitive performance is particularly evident in the US101 case because traffic evolution in this dataset is dominated by the continuous propagation of congestion waves rather than isolated pointwise fluctuations. Compared with the positional encoding in Transformers, the operator-learning mechanism of STFNO is better aligned with such traffic-flow dynamics, as it models the traffic state as a continuous field and captures long-range spatiotemporal dependencies through global spectral interactions.

Fig. 4 showcases the speed field reconstruction heatmaps for STFNO applied to the US101 dataset, depicting results based on 10%, 20%, 30%, 40%, and 50% available detection data, with the final subplot representing the observed global speed field. Remarkably, training STFNO with only 10% of the available observations yields a highly effective speed field reconstruction, as evidenced by the coherent patterns in the heatmap. As the proportion of available detection data increases, the reconstruction quality improves noticeably, with the fields becoming more precise and closely aligned with the true global observation, particularly evident in the 40% and 50% scenarios where the patterns sharpen and stabilize. Furthermore, STFNO excels at capturing global features, effectively learning the mapping between function spaces to reconstruct the evolution of traffic flow shocks and sparse waves with high fidelity. This capability stems from its operation in function space, leveraging Fourier transforms to map the flow field into the frequency domain, enabling efficient global modeling with low-dimensional parameters that capture critical physical patterns, thus achieving accurate traffic speed field reconstruction.

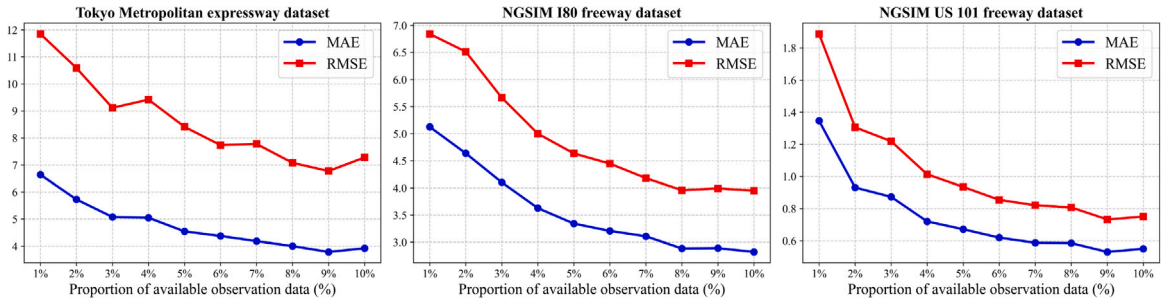


Fig. 5. Performance evolution of STFNO under highly sparse scenarios across three datasets.

Table 4
Comparison of model complexity (parameters number, memory usage and computational speed).

Model	Parameter	NGSIM US 101 freeway dataset		NGSIM I80 freeway dataset		Tokyo Metropolitan expressway dataset	
		Memory/MB	Time/s	Memory/MB	Time/s	Memory/MB	Time/s
NN	8577	17.47	0.24	16.96	0.24	17.32	0.29
LSTM	117 057	23.71	1.82	23.19	0.96	23.43	1.48
Transformer	550 113	25.76	11.8	25.24	6.12	25.58	9.64
TimeMixerPP	156 549	18.65	30.27	18.70	29.47	19.10	28.75
STFNO	177 673	21.52	0.87	21.52	0.46	22.06	0.75

Additionally, Fig. 5 quantitatively illustrates the performance decay of STFNO as the observation rate decreases from 1% to 10% on the three datasets. Overall, both MAE and RMSE increase as the available observation data become sparser, which is consistent with the growing difficulty of traffic state reconstruction under extremely limited sensing conditions. Nevertheless, STFNO still maintains a reasonable reconstruction capability even in highly sparse scenarios. The curves suggest that when the observation rate is around 5% and 6% or higher, STFNO can still maintain relatively stable reconstruction performance, whereas below this range the errors tend to increase more noticeably.

4.3.2. Model complexity of STFNO and baseline models

Table 4 provides an evaluation of model complexity across the NGSIM US101, NGSIM I80, and Tokyo datasets, based on 50% available detection data, with computational efficiency measured over 100 epochs. The times in the table are the average times required for one training epoch. STFNO, with 177673 parameters, exceeds NN, LSTM and TimeMixerPP, but remains significantly lighter than the Transformer, offering a balance of high performance and reduced parameter overhead. In terms of memory usage, STFNO requires 21.52 MB for US101, 21.52 MB for I80, and 22.06 MB for Tokyo, which is lower than that of LSTM and Transformer, although slightly higher than that of NN and TimeMixerPP, which enables efficient deployment on limited hardware. Computationally, STFNO excels with rapid processing, notably achieving 0.46 s for I80, enhancing efficiency compared to 6.12 s of Transformer and 29.47 s of TimeMixerPP. Through low-frequency modal learning in the frequency domain combined with spatial residual connections, STFNO achieves a significant breakthrough in balancing model complexity, computational efficiency, and reconstruction accuracy. Additionally, STFNO employs low-rank decomposition with a rank of 0.4 here for model training, suggesting potential for further reductions in parameter count, memory usage, and runtime without notable performance degradation.

4.4. Results: zero-shot super-resolution TSR

Compared to the aforementioned method of relying on low-resolution data to capture the dynamic characteristics of traffic waves and then reconstructing the traffic field at a fixed resolution, the more important thing is to generalize traffic field reconstruction across resolutions. This facilitates high-precision reproduction of key traffic dynamics without resolution restrictions. Therefore, we conducted experiments on temporal generalization, spatial generalization, and spatiotemporal generalization on three datasets to evaluate and verify the generalization performance of STFNO.

4.4.1. Temporal generalization on Tokyo Metropolitan expressway case

Table 5 details the generalized reconstruction accuracy of STFNO and baseline models trained on low-resolution 23×168 scenarios with 50%, 70%, and 90% available detected data, then generalized to high-resolution 23×2016 fields for testing—a 12-fold increase in temporal resolution with spatial dimension held at 23. The models, having converged during training, are tested using only spatiotemporal coordinates as input for the high-resolution reconstruction. STFNO demonstrates a significant generalization advantage across all sparsity levels. With 50% available detected data, STFNO achieves an MAE of 3.48 and RMSE of 6.11, with improvements of 3.87% and 4.23% over all baselines. This advantage persists at 70%, with STFNO recording an MAE of 3.16 and RMSE of 5.38, compared with 3.43 and 5.96 for TimeMixerPP. At 90% sparsity, STFNO's MAE of 3.04 and RMSE of 5.02

Table 5
Accuracy comparison of zero-shot super-resolution speed field reconstruction between STFNO and baselines trained with different proportion of observation data in the Tokyo Metropolitan expressway case (12 times finer resolution).

Model	50%		70%		90%	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
NN	4.89	7.62	4.86	7.76	4.65	7.25
LSTM	4.14	7.02	4.21	7.15	3.94	6.51
Transformer	3.97	7.32	3.70	6.55	3.58	6.30
TimeMixerPP	3.62	6.38	3.43	5.96	3.26	5.67
STFNO	**3.48**	**6.11**	**3.16**	**5.38**	**3.04**	**5.02**
Gain	3.87%	4.23%	7.87%	9.73%	6.75%	11.46%

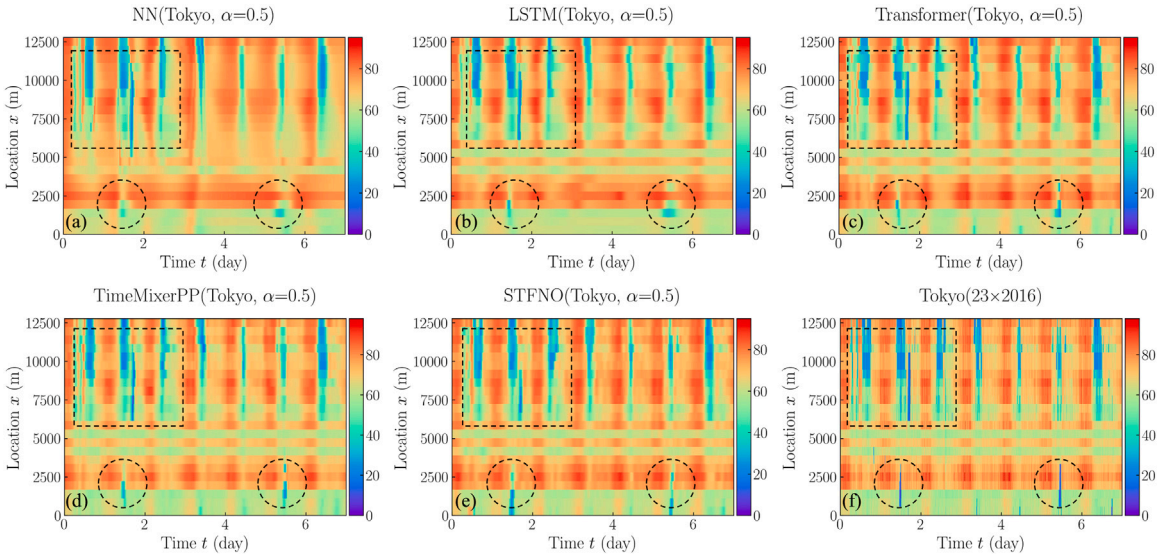


Fig. 6. Comparison of high-resolution speed field reconstruction in the temporal generalization of different models trained with 50% available low-resolution detected data in the Tokyo expressway case.

yield further improvements over the best baseline. This superior performance stems from the ability of STFNO to learn operators within continuous function spaces, leveraging its function-to-function mapping approach to effectively capture the underlying traffic dynamics across low and high resolutions.

Fig. 6 presents the heatmaps of high-resolution speed field reconstruction for STFNO and baseline models trained with 50% available low-resolution detected data in the Tokyo expressway case, showcasing their temporal generalization performance. Overall, STFNO delivers the most accurate reconstruction of the global speed field, with patterns closely mirroring the true dynamics. The black dashed rectangles highlight recurring drops in traffic speed, corresponding to increased traffic loads during morning and evening peak hours, where STFNO's ability to capture and learn the evolution of these traffic waves significantly surpasses the baseline models. Additionally, the black dashed circles mark irregular congestion on specific road segments, which STFNO effectively reconstructs with greater detail. The superior performance aligns with the design of STFNO, as the macroscopic patterns of traffic speed field (such as congestion bands and free-flow regions) are mainly dominated by low-frequency components, which is a perfect fit with the architecture of STFNO.

Fig. 7 illustrates the sample fit comparison between STFNO and benchmark models, with observed traffic speeds on the x-axis and reconstructed high-resolution speed field values on the y-axis, where proximity to the diagonal line indicates higher accuracy. The dashed boxes highlight that sample fitting degree of STFNO is markedly superior, with data points clustering more tightly around the diagonal compared to the baselines, reflecting a closer alignment. The reconstructed sample distribution of STFNO also more closely matches the observed distribution. This enhanced performance is likely due to the fact that FNO uses Fourier transforms to map time series into the frequency domain, enabling automatic identification of inherent periodic patterns in traffic flow. This frequency-domain representation allows the model to separate steady-state patterns from transient fluctuations, facilitating better generalization to finer-grained time periods at high resolution.

4.4.2. Spatial generalization on NGSIM I80 freeway case

Table 6 shifts focus to spatial generalization using the I80 dataset, contrasting with the previous temporal generalization in Table 5, where STFNO and baselines are still trained with 50%, 70%, and 90% available low-resolution detected data for a fair

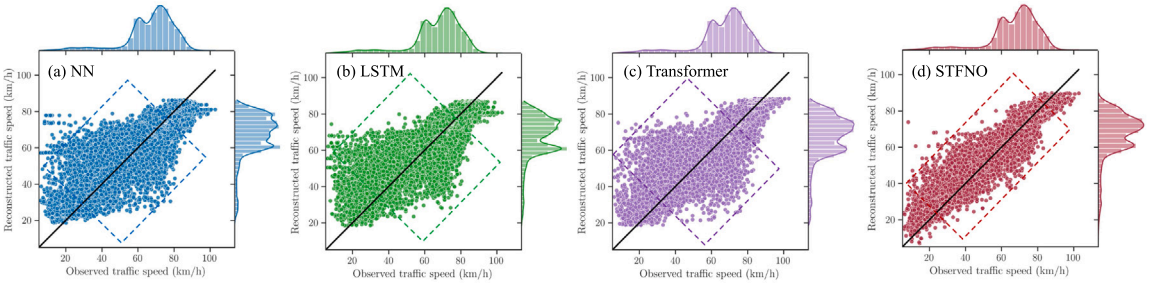


Fig. 7. Comparison of sample fit between STFNO and the baseline models when generalized to high-resolution fields with 90% low-resolution available detected data in the Tokyo expressway case.

Table 6
Accuracy comparison of zero-shot super-resolution speed field reconstruction between STFNO and baselines trained with different proportion of observation data in the NGSIM I80 case (9 times finer resolution).

Model	50%		70%		90%	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
NN	3.34	4.58	3.14	4.35	2.55	3.42
LSTM	3.13	4.26	2.72	3.86	2.64	3.59
Transformer	2.87	3.97	2.69	3.77	2.62	3.56
TimeMixerPP	2.97	4.04	2.63	3.63	2.53	3.43
STFNO	**2.77**	**3.88**	**2.51**	**3.55**	**2.36**	**3.32**
Gain	3.48%	2.27%	4.56%	2.20%	6.72%	2.92%

comparison under sufficiently trained conditions. STFNO consistently outperforms. With 50% low-resolution training data, STFNO achieves a MAE of 2.77 and an RMSE of 3.88. This advantage remains stable with 90% data, where it delivers 2.36 MAE and 3.32 RMSE, compared with 2.53 and 3.43 for TimeMixerPP, underscoring its superior spatial generalization capability in speed reconstruction tasks. This result suggests that, under spatial upsampling, STFNO is better able to preserve field-level spatial continuity than sequence-based baselines that remain more tied to the training grid resolution.

Fig. 8 plots the heatmaps of high-resolution speed field reconstruction for spatial generalization, with models trained on low-resolution 9 × 360 data and generalized to a 9-fold spatially refined 81 × 360 field using 90% available detected data, where optimal performance has been achieved. The contrast is striking: as shown in subfigures Figs. 8(a)–(d), while NN, LSTM, Transformer and TimeMixerPP exhibit poor performance post-generalization, failing to reconstruct even shallow trends and patterns, STFNO maintains its effectiveness. More specifically, NN, LSTM, and Transformer fail to recover the narrow stripe-like traffic structures after spatial refinement, and their reconstructed fields are overly smoothed. TimeMixerPP preserves part of the global trend, but its reconstructed field still appears blurred and misses some localized structures. As marked by the black dashed circles in subfigures Figs. 8(e) and (f), STFNO not only accurately reconstructs the global traffic flow evolution trends but also captures detailed traffic wave dynamics. In particular, the localized sharp structure near $t = 900$ s s is much better recovered by STFNO than by the baseline models, indicating that STFNO can more effectively preserve both large-scale propagation patterns and local spatial details during cross-resolution reconstruction. This spatial generalization advantage may be due to the translation invariance and discretization invariance of STFNO. As a mapping between function spaces, the learning process of STFNO is decoupled from the specific grid resolution.

Fig. 9 presents the sample fit comparison for STFNO and baseline models, with observed traffic speeds on the x-axis and reconstructed high-resolution speeds on the y-axis using 50% low-resolution detected data in the NGSIM I-80 freeway case, where proximity to the diagonal indicates better accuracy. For speeds above 40 ft/s, baseline models exhibit significant deviations, as highlighted by the dashed boxes, with reconstructed values clustering around 20 ft/s. In contrast, STFNO shows smaller deviations, with its scattered points, even when off-target, remaining sparsely distributed around the diagonal, underscoring its superior fitting capability. It is also worth noting that a visible spread still appears around the observed speed of about 20 ft/s. A possible reason is that this range may correspond to traffic transition states, where local patterns are more heterogeneous, so under cross-resolution reconstruction small mismatches may be amplified into noticeable pointwise deviations.

4.4.3. Spatiotemporal generalization on NGSIM US 101 freeway case

Table 7 presents the spatiotemporal generalization results, transitioning from a low-resolution 8 × 180 field to a 39-fold higher-resolution 104 × 540 field. STFNO outperforms baselines across 50%, 70%, and 90% available detected data, achieving an MAE of 0.84 and RMSE of 1.14 with 50% training low-resolution data, marking a 16.83% MAE and 16.79% RMSE gain over the Transformer. At this observation level, STFNO also clearly outperforms TimeMixerPP, whose MAE and RMSE are 1.20 and 1.58, respectively. This advantage grows with 70% available detected data, where STFNO records an MAE of 0.72 and RMSE of 0.97, offering a 26.53%

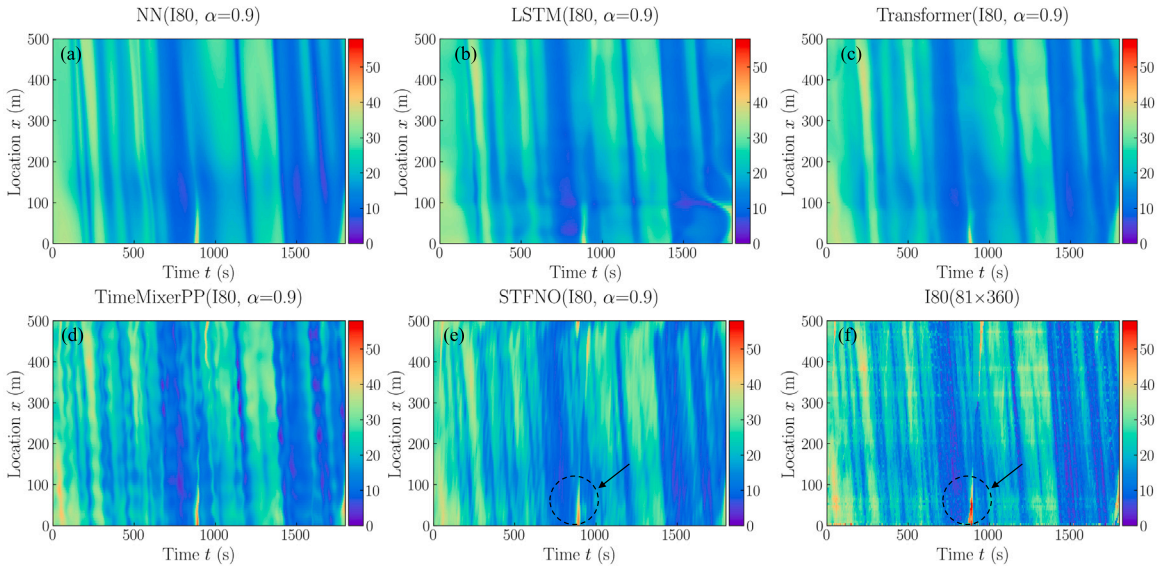


Fig. 8. Comparison of high-resolution speed field reconstruction in the spatial generalization of different models trained with 90% available low-resolution detected data in the NGSIM I80 freeway case.

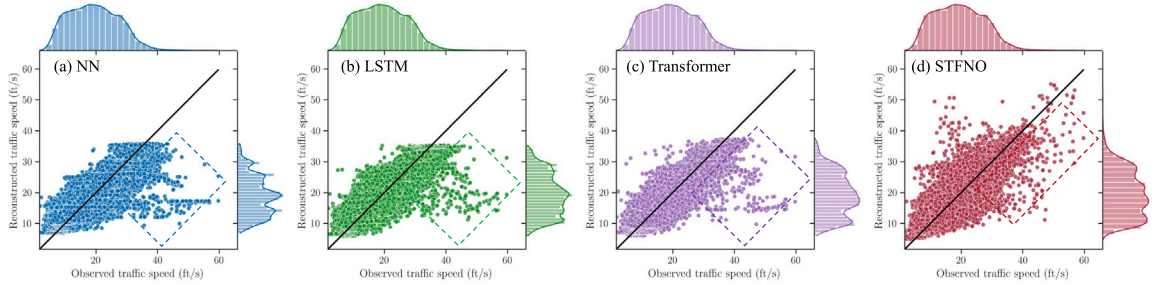


Fig. 9. Comparison of sample fit between STFNO and the baseline models when generalized to high-resolution fields with 50% low-resolution available detected data in the NGSIM I80 freeway case.

Table 7
Accuracy comparison of zero-shot super-resolution speed field reconstruction between STFNO and baselines trained with different proportion of observation data in the NGSIM US101 case (39 times finer resolution).

Model	50%		70%		90%	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
NN	1.07	2.43	1.11	1.53	0.85	1.11
LSTM	1.46	2.01	0.98	1.36	0.89	1.18
Transformer	1.01	1.37	0.98	1.32	0.80	1.05
TimeMixerPP	1.20	1.58	1.37	1.50	1.02	1.36
STFNO	**0.84**	**1.14**	**0.72**	**0.97**	**0.62**	**0.83**
Gain	16.83%	16.79%	26.53%	26.52%	22.5%	20.95%

MAE and 26.52% RMSE improvement. At 90% training data, STFNO delivers an MAE of 0.62 and RMSE of 0.83, with gains of 22.5% and 20.95%, respectively, highlighting its robust accuracy in complex spatiotemporal speed field reconstructions. This result is closely related to the traffic characteristics of the US101 corridor, where shockwave propagation, queue growth, and congestion dissipation evolve continuously across both space and time. Since STFNO learns an operator over the traffic field rather than a resolution-specific sequence representation, it is better able to preserve these cross-scale traffic dynamics when generalized from low-resolution observations to a much finer spatiotemporal grid.

Fig. 10 displays the heatmaps of speed field reconstruction in the US101 dataset, highlighting the generalization performance of STFNO over the baselines in capturing global patterns. Figs. 10(a)–(c) for NN, LSTM, and Transformer show overly smoothed details, as their training is often limited by data discretization, resulting in poor generalization across different grid resolutions and

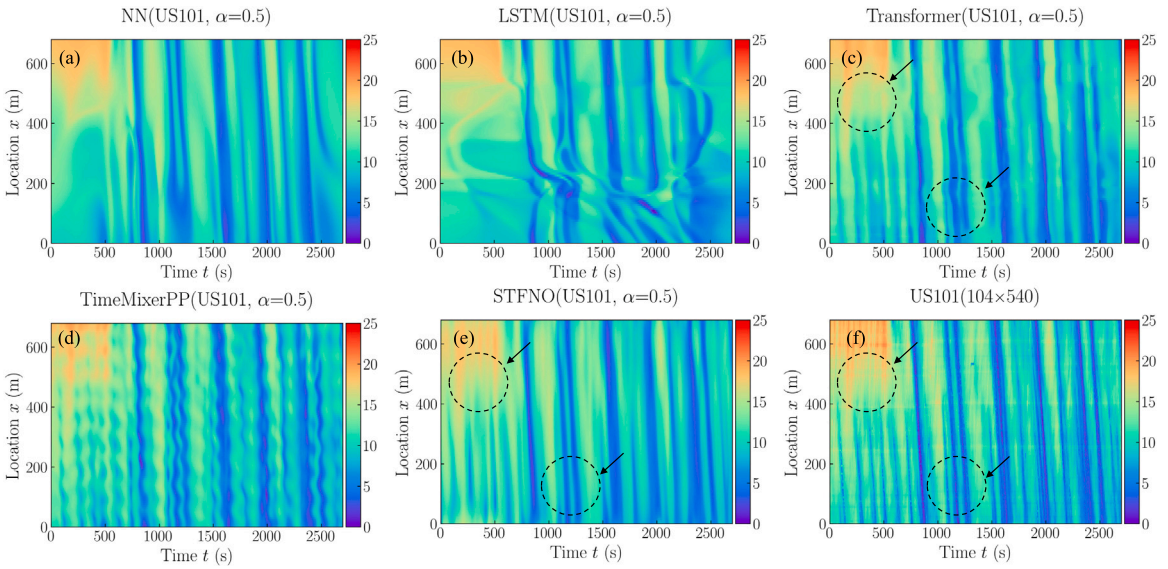


Fig. 10. Comparison of high-resolution speed field reconstruction in the spatiotemporal generalization of different models trained with 50% available low-resolution detected data in the NGSIM US101 freeway case.

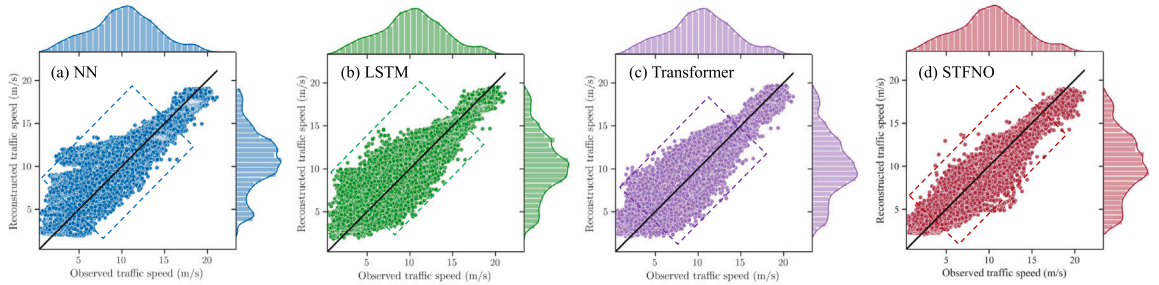


Fig. 11. Comparison of sample fit between STFNO and the baseline models when generalized to high-resolution fields with 70% low-resolution available detected data in the NGSIM US101 freeway case (39 times finer resolution).

failing to accurately reflect the continuous propagation of congestion waves. TimeMixerPP in Fig. 10(d) preserves the global trend to some extent, but its reconstructed field still exhibits noticeable smoothing and fails to recover key localized structures as accurately as STFNO. Notably, the upper-left high-speed free-flow region is blurred in these models, whereas STFNO precisely delineates its evolution, as shown in Fig. 10(e). While STFNO does not achieve perfect high-precision reconstruction, it more accurately captures the stable traffic characteristics up to around 1000 s and the subsequent shock waves relative to the baselines.

Fig. 11 illustrates the sample fit comparison for STFNO and baseline models trained with 70% dataset, where proximity to the diagonal indicates accuracy. Baseline models tend to overestimate speeds, with significant deviations evident at around 5 m/s, where reconstructed samples show large errors. STFNO, while also exhibiting notable deviations in the 10–15 m/s range, demonstrates a tighter distribution of points around the diagonal, as highlighted by the dashed boxes, compared to the broader scatter of baseline models. Trained on low-resolution data and generalized to high-resolution speed field reconstruction, STFNO proves effective and superior. This advantage arises from its resolution-independent continuity, mapping function spaces to function spaces, rendering STFNO robust to changes in spatiotemporal resolution-whether in temporal sampling rates or spatial discretization-ensuring the learned operator remains effective.

5. Conclusion

This study proposes an end-to-end traffic state reconstruction (TSR) framework that breaks through the resolution limitations inherent in previous TSR methods. The framework utilizes the spatio-temporal Fourier neural operator (STFNO) as its core method, capturing the essential laws of traffic flow through operator learning in the frequency domain, thus overcoming the limitations of traditional neural networks in processing discrete grid data. Considering that actual traffic flow data often exhibits sparsity and low resolution, this paper sets up TSR tasks under both fixed-resolution and super-resolution conditions based on datasets from multiple

Table A.1
Accuracy comparison of zero-shot super-resolution speed field reconstruction between STFNO and baselines trained with different proportion of observation data in the NGSIM US101 case (312 times finer resolution).

Model	50%		70%		90%	
	MAE	RMSE	MAE	RMSE	MAE	RMSE
NN	2.56	3.44	1.70	2.21	3.14	4.13
LSTM	2.51	3.37	2.14	2.80	1.90	2.54
Transformer	2.34	3.03	2.06	2.65	1.78	2.27
TimeMixerPP	2.42	2.98	1.98	2.45	2.08	2.55
STFNO	**1.78**	**2.31**	**1.62**	**2.15**	**1.39**	**1.78**
Gain	23.93%	22.48%	4.71%	2.71%	21.91%	21.59%

real-world scenarios. In the fixed-resolution task, STFNO demonstrates competitive reconstruction performance across different sampling rates, while also showing advantages in terms of computational complexity and potential for further optimization. In the zero-shot super-resolution task, benefiting from the resolution invariance of the neural operator, STFNO achieves significantly better performance than the baseline model in temporal generalization, spatial generalization, and spatiotemporal joint generalization, with MAE improvements ranging from 3.48% to 26.53% and RMSE improvements ranging from 2.20% to 26.52% across different sparse scenarios. Furthermore, it significantly surpasses existing models in reconstructing key traffic flow phenomena such as stop-and-go waves and shock waves. These results demonstrate the superiority and effectiveness of the proposed framework in TSR, effectively addressing the limitations of sparse and coarse-grained data in TSR.

Future work will focus on several directions closely related to the limitations of this study. First, combining operator learning with dynamic graph-based modeling is a promising direction for handling non-Euclidean traffic systems and irregular sensor layouts. Second, explicit traffic-flow physical laws can be incorporated into the operator-learning process to further improve the physical consistency of reconstructed traffic fields. Third, since single-source data was used in this paper, multi-source data fusion [50,51] is worth further exploration for more comprehensive traffic state reconstruction. Furthermore, the potential of large language models in traffic modeling [52] may also be explored for more flexible integration of heterogeneous traffic information.

CRediT authorship contribution statement

Ting Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Zhihao Li:** Visualization, Validation, Methodology, Data curation, Conceptualization. **Rongjun Cheng:** Writing – review & editing, Writing – original draft, Supervision, Funding acquisition. **Ling Guo:** Supervision, Methodology, Conceptualization. **Takao Dantsuji:** Supervision, Data curation. **Ye Li:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Extreme spatiotemporal generalization scenario

Table A.1, Figs. A.1 and A.2 present results for an extreme spatiotemporal generalization scenario, where models are trained on a low-resolution 4×45 field and generalized to a 104×540 high-resolution field, achieving a 312-fold refinement for speed field reconstruction in the NGSIM US101 case. STFNO consistently outperforms baselines across various available detected data proportion, with gains such as 23.93% MAE and 22.48% RMSE at 50%, and 21.91% MAE and 21.59% RMSE at 90%. As shown in heatmap Fig. A.1, while STFNO does not achieve precise reconstruction, it significantly outperforms the baseline models that fail entirely, marking a notable breakthrough in generalization. Similarly, the scatter plots Fig. A.2 reveal the effectiveness of STFNO in this challenging scenario, with points more closely aligned around the diagonal compared to the baselines.

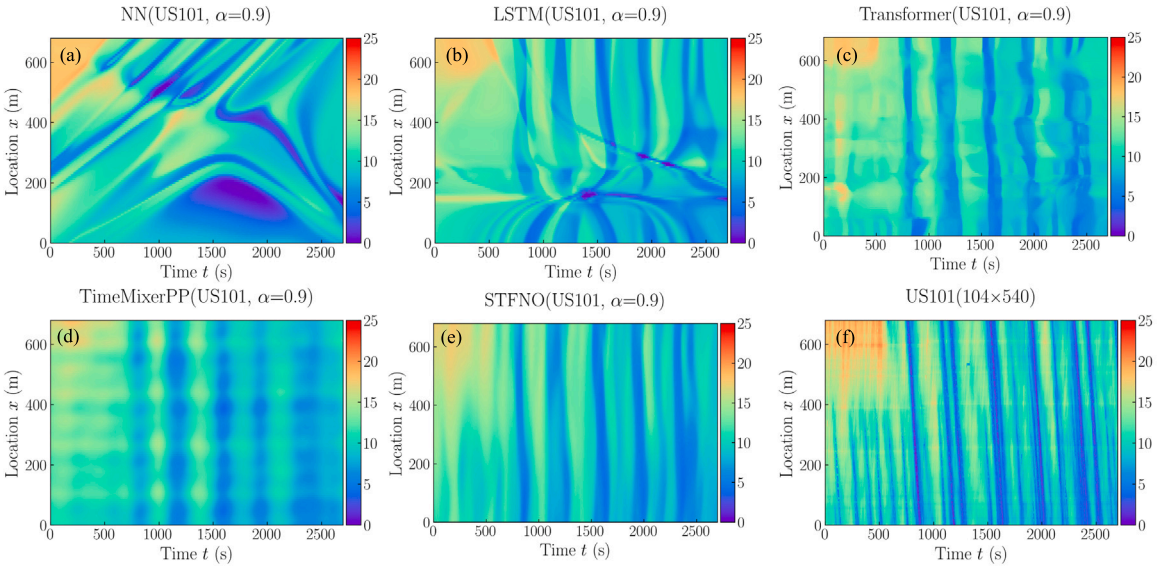


Fig. A.1. Comparison of high-resolution speed field reconstruction in the spatiotemporal generalization of different models trained with 90% available low-resolution detected data in the NGSIM US101 freeway case (312 times finer resolution).

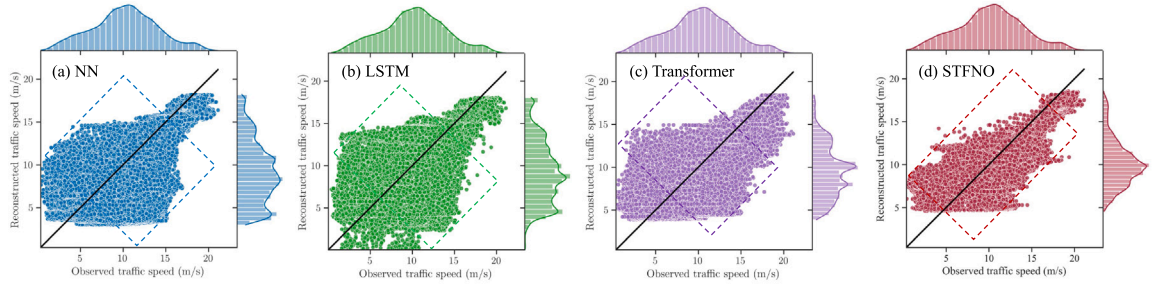


Fig. A.2. Comparison of sample fit between STFNO and the baseline models when generalized to high-resolution fields with 50% low-resolution available detected data in the NGSIM US101 freeway case (312 times finer resolution).

Potential model extensions using NFFT

For more general traffic sensing scenarios, detector measurements may be collected at irregular spatial locations and cannot always be naturally aligned with a uniform reference grid. A possible extension is to replace the FFT operations in the spectral branch with NFFT-based transforms, thereby generalizing the present framework toward a nonequispaced Fourier neural operator. Let $\xi_p = (x_p, t_p)$ denote an irregular observation coordinate. Then the spectral transform can be expressed in the form:

$$\hat{\mathbf{H}}(k) = \sum_{p=1}^{P} \mathbf{H}(\xi_p)\, e^{-i2\pi k \cdot \xi_p} \tag{A.1}$$

where P is the number of observed points and k denotes the spatiotemporal frequency index.

Under this extension, the current dual-path design can still be preserved. The spectral branch is used to capture global spatiotemporal dependencies from irregular observations, while the spatial shortcut branch continues to retain local traffic details. The layer output may still be written as:

$$\mathbf{H}_{l+1}(\xi) = \sigma\left[V_l(\xi) + S_l(\xi)\right] \tag{A.2}$$

In this way, the model could be generalized beyond grid-aligned inputs while still supporting super-resolution reconstruction on a target high-resolution grid. This extension is not implemented in the present study, but it provides a possible direction for improving the practical applicability of STFNO under fully irregular sensing conditions.

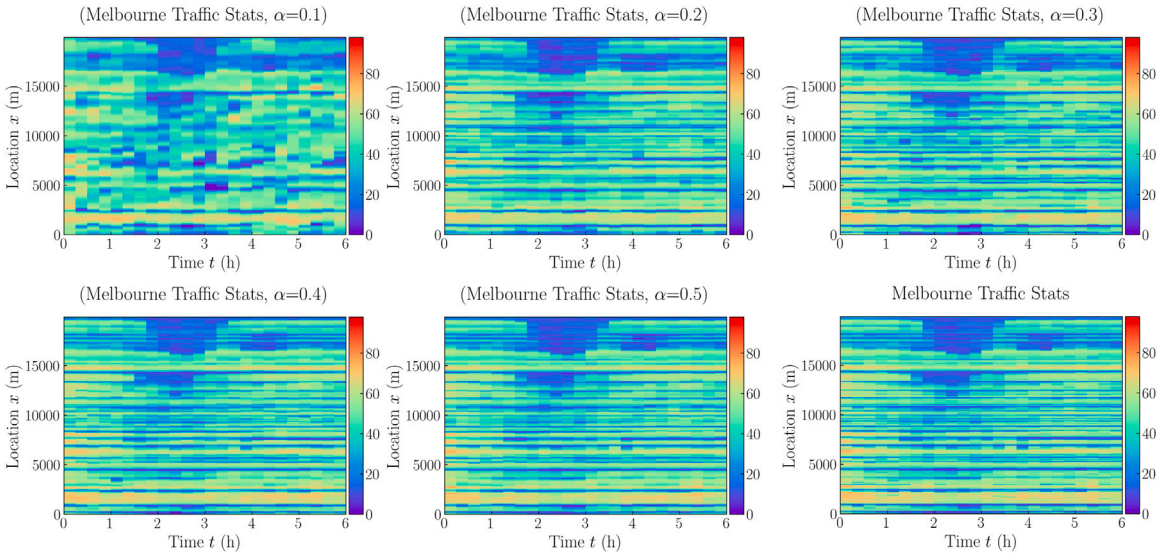


Fig. A.3. Heatmap of speed field reconstruction by STFNO under Melbourne Traffic Stats dataset.

Traffic state field reconstruction in urban scenarios

In addition to the freeway scenario, we supplemented the datasets with a Melbourne Traffic Stats dataset[1] for urban traffic scenarios. This dataset was collected from a long corridor road in Melbourne from Wellington Street to Flinders Street,[2] with a total length of 19.91 km and divided into 485 segments, spanning from 6:00 AM to 12:00 PM on June 12, 2023. The reconstruction results by STFNO are shown in Fig. A.3, which also demonstrates the effectiveness of STFNO in traffic state field reconstruction.

Data availability

Data will be made available on request.

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